

An Applicant's Guide To USPTO's Examiner Incentive System

By **Mandy Wilson Decker** (February 8, 2023)

Incentives are powerful tools for driving behavior. On this topic, billionaire businessman Charlie Munger famously advised to "[n]ever, ever, think about something else when you should be thinking about the power of incentives." [1]

Because the pathway from patent application to granted patent depends heavily on the actions and judgments of the examiner assigned to the application, a patent applicant is well advised to consider the examiner's perspective and the incentives imposed by the U.S. Patent and Trademark Office through the examiner production system, also referred to as the count system.



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The count system is somewhat complex. The goal of this article is to provide the reader with a guide for understanding the system, in order to recognize whether particular behaviors are directly or indirectly incentivized — or disincentivized — by the system.

Such knowledge can be leveraged to improve patent protection strategies. The current economy is characterized by high inflation, high demand for labor, and high competition for innovative products and services.

Improving patent strategies not only facilitates more rapid acquisition of intellectual property assets, but also injects cost savings into the acquisition process, which can be deployed for other critical needs.

Examiner Annual Performance Rating

Patent examiners' annual performance rating is based on a weighted average of four elements: [2]

1. Productivity (Production) — 35%
2. Quality — 35%
3. Docket management — 20%
4. Stakeholder interaction — 10%

Together, production, meaning the number of production units, and docket management, meaning the completion of those PUs within an expected time frame, capture 55% of the weighted average. Examiners are also evaluated based on the quality [3] of their work and interactions with internal and external contacts.

This article focuses on the production and the docket management elements of the annual performance rating.

Formula for Expected Production

The following formula is used to calculate expected production for a patent examiner:

$$\frac{(\text{Number of Examining Hours}) \times (\text{Seniority Factor})}{(\text{Technology Complexity})} = \text{Production Units (PU)}$$

Seniority Factor

Patent examiners are civilian federal employees with seniority grade levels as defined by the general schedule, or GS, classification system.[4]

The grade level affects an examiner's salary and, though the seniority factor assigned by the USPTO, it also affects the number of production units required during a given time period.

Most examiners start as a GS-7 or GS-9,[5] but could be classified at GS-5 to GS-14. GS-5 to GS-12 examiners do not have signatory authority, and their office actions must be approved by a primary examiner or a supervisory patent examiner.[6]

GS-13 and GS-14 examiners can become eligible for the partial signatory authority program, and after an additional testing phase, a GS-14 examiner can become a primary examiner with full signatory authority.[7] If an examiner is promoted to supervisory patent examiner at GS-15, the primary role shifts to that of supervising other patent examiners.[8]

Table 1 identifies the seniority factors assigned by the USPTO for examiners based on their GS grade level.[9]

Table 1: Seniority Factors for Different Examiner Grade Levels

Examiner Grade Level	Seniority Factor
GS-5	0.55
GS-7	0.7
GS-9	0.8
GS-11	0.9
GS-12	1.0
GS-13	1.15
GS-13/ -14 with PSA	1.25
GS-14 with FSA	1.35

Number of Examining Hours

Full-time patent examiners work 40 hours a week,[10] but not all of this work time is allocated to examining hours.

For example, examining hours exclude training, staff meetings and certain programs where examiners receive additional time, such as in connection with the After Final Consideration Pilot 2.0 program.[11] Thus, a typical number of examining hours is provided in Table 2.[12]

**Table 2: Weekly and Bi-Weekly Hours for
Full Time Examiners**

	Total Hours	Non- Examining Hours	Examining Hours
Weekly	40	4	36
Bi-Weekly Pay Period	80	8	72

Technical Complexity

For purposes of the formula for expected production, the technical complexity number is provided in units of hours per production unit. Generally, patent applications directed to technologies of greater complexity will be assigned a greater number of hours per PU because there is an expectation that additional time will be required to examine more complex applications.

Baseline expectancies range from 13.8 to 31.6 hours per PU and are identified with consideration to a technology classification system known as the Cooperative Patent Classification system.[13]

For example, baseline for an application directed to a fishing lure could be 16.6 hours per PU, baseline for an immunotherapy application could be 25.9 hours per PU, and baseline for a satellite communication application could be 27.7 hours per PU.[14]

There is ongoing work at the USPTO to refine appropriate examination times, with consideration to factors affecting individual applications, such as claim number and information disclosure statement volume.[15]

Examples: Applying the Formula for Expected Production

In Table 3, the formula is applied for a biweekly pay period to four hypothetical examiners with distinct seniority factors based on their grade levels, and examining cases from distinct technology areas.

Table 3: Examples Illustrating Expected PU and Hours/PU for Examiners in Different Grade Levels Working in Different Technology Areas

Examiner	Seniority Factor	Exam. Hours	Tech Area	Technical Complexity (Hrs/PU)	Expected Production Units (PU)	Hours/ PU
GS-7	0.7	72	Immuno-therapy	25.9	$\frac{(0.7)(72)}{25.9} = 1.9$	$\frac{72}{1.9} = 37.9$
GS-7	0.7	72	Fishing Lure	16.6	$\frac{(0.7)(72)}{16.6} = 3.0$	$\frac{72}{3.0} = 24.0$
GS-14 w/ FSA	1.35	72	Immuno-therapy	25.9	$\frac{(1.35)(72)}{25.9} = 3.8$	$\frac{72}{3.8} = 18.9$
GS-14 w/ FSA	1.35	72	Fishing Lure	16.6	$\frac{(1.35)(72)}{16.6} = 5.9$	$\frac{72}{5.9} = 12.2$

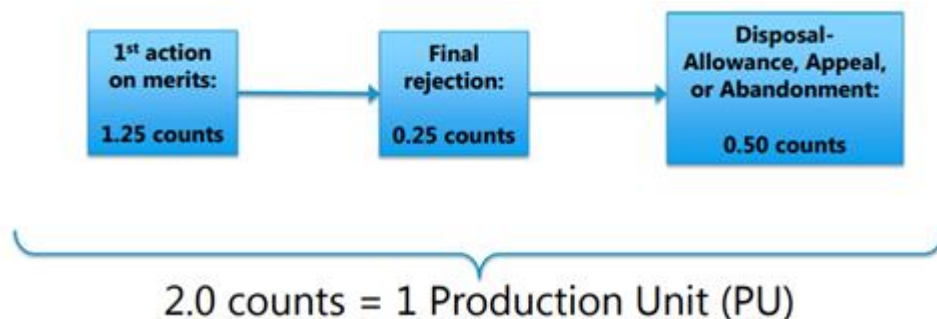
The information in Table 3 illustrates the following:

- For examiners of the same grade level, there is a lower number of expected production units when working in a more complex technology area, like 1.9 for GS-7 in immunotherapy, as compared to a less complex technology area, such as 3.0 for GS-7 fishing lures.
- A less experienced[16] examiner, like a GS-7, will have a lower number of expected production units — 1.9 for immunotherapy — as compared to a more experienced examiner like a GS-14 in the same technology area, who would have 3.8 for immunotherapy.
- A less experienced examiner like a GS-7 has a greater number of hours allocated to each PU — 37.9 for immunotherapy — as compared to a more-experienced examiner like GS-14, who would have 18.9 for immunotherapy.

To understand how an examiner would operate to meet these expected production levels, a production unit needs to be defined.

Production Unit

One production unit equals two counts, with counts being awarded for certain actions taken during a prosecution cycle, as illustrated in the following figure from the USPTO:[17]



For a given patent application, a greater number of counts are assigned to the first office action on the merits, as compared to the remaining actions in the prosecution cycle.

As illustrated in the figure above, for a given application, a prosecution cycle includes two office actions and a disposal.

However, most examiners have an office action per grant ratio[18] that is greater than two, and for certain technology areas the office action per grant ratio is notably greater than two. Where three or more office actions are issued for a given application, the applicant has typically filed a request for continued examination, compelling the examiner to take action beyond the original prosecution cycle.

In addition to counts associated with actions in the original prosecution cycle, Table 4 includes counts for applications involving requests for continued examination[19], and translates these counts into hours for GS-7 and GS-14 examiners working on immunotherapy technologies:

Table 4: Counts and Hours Allocated to Actions in Original and RCE Prosecution Cycles for Immunotherapy Patent Applications for Examiners of Different Grade Levels

	Original Prosecution Cycle			1 st RCE			2 nd & Subsequent RCEs		
	FAOM	Final	Disposal	FAOM	Final	Disposal	FAOM	Final	Disposal
Counts	1.25	0.25	0.5	1.0	0.25	0.5	0.75	0.25	0.5
	Total: 2.0			Total: 1.75			Total: 1.50		
GS-7 Hours¹	23.6	4.7	9.5	18.9	4.7	9.5	14.2	4.7	9.5
	Total: 37.8			Total: 33.1			Total: 28.4		
GS-14 Hours²	11.9	2.4	4.8	9.5	2.4	4.8	7.1	2.4	4.8
	Total: 19.1			Total: 16.7			Total: 14.3		

¹ (72 hours)/(1.9 PU) x (1 PU)/(2 counts) = 18.9 hours per count

² (72 hours)/(3.8 PU) x (1 PU)/(2 counts) = 9.5 hours per count

The information in Table 4 demonstrates the following:

- Examiners have defined time constraints for actions during each phase of a prosecution cycle.

- Examiners have a greater amount of time allocated to the beginning of each prosecution cycle, i.e., the time leading up to and including the issuance of a first office action on the merits, with the most time being allocated during the original prosecution cycle.
- Grade-level promotions are tied to a reduction in examination time allocated to each patent application, with promotion from GS-7 to GS-14 results in the examination time being cut by about 50%.

Production Measured in Counts, Not Time

Although there are a certain number of hours an examiner is allocated to complete actions during a prosecution cycle, the examiner's production is not evaluated based on actual hours spent.

Rather, the examiner's production is evaluated based on PUs or counts. In other words, production evaluation depends on the number of office actions issued, not the time spent to issue those office actions.

Indeed, spending less than the allocated time on an office action is contemplated and encouraged, as evidenced by a USPTO production award program that awards examiners for exceeding 100% production.[20] A production award program is summarized in the following figure from the USPTO:[21]

Production	110%	115%	120%	125%	130%	135%
Award	2%	3%	4%	5%	6%	7%

In Table 5, the decrease in the amount of hours per count needed to achieve 135% of expected production is provided for the four hypothetical examiners from Table 3.

Table 5: Decrease in Hours per Count with Production in Excess of 100% Goal

Examiner	Tech Area	100% Expected PU	100% Expected Counts	135% Expected Counts	Hours per Count at 100%*	Hours per Count at 135%*
GS-7	Immuno-therapy	1.9	3.8	5.1	18.9	14.1
GS-7	Fishing Lure	3.0	6.0	8.1	12.0	8.9
GS-14 w/ FSA	Immuno-therapy	3.8	7.6	10.3	9.5	7.0
GS-14 w/ FSA	Fishing Lure	5.9	11.8	15.9	6.1	4.5
*Assuming 72 examining hours per bi-weekly pay period, without overtime.						

The calculations from Table 5, in view of the production award program, indicated that:

- Examiners are encouraged to exceed 100% production.
- Increasing production from 100% to 135% involves a reduction in time of about 25% to obtain each count.

Strategies to Employ in View of the Count System

Experienced applicants will have many examples of instances when examiners took action seemingly based only on an intrinsic desire to perform their job well. Nevertheless, from a rational[22] perspective, the following actions would tend to maximize performance rating and compensation under the count system:

- Allocate a greater amount of time to the initial prosecution cycle and to actions at the beginning of each prosecution cycle.
- Minimize time spent on actions within each prosecution cycle whenever possible to increase overall production metrics.
- Consider opportunities for additional time under USPTO programs.
- Pursue promotions in seniority grade levels within the GS classification system to obtain compensation increases, while accepting the associated reduction in examination time allocated to each patent application.

From this perspective, patent protection strategies can be improved by taking steps to minimize the time necessary for the examiner to fully examine and allow a patent application. To this end, the following are a few examples of steps that could be taken:

- Consider preliminary amendments to claims prior to examination to best define the scope of the invention in view of currently preferred commercial embodiments and in view of U.S. law. This consideration may be particularly useful when there is a lengthy pendency prior to a first office action on the merits or when the application was initially drafted for filing in a jurisdiction other than the U.S.
- Engage meaningfully with the examiner at the beginning of the initial prosecution cycle, including providing a thoughtful and complete response to the first office action on the merits.
- Where appropriate, make use of After Final Consideration Pilot 2.0, which provides the examiner with some additional time.
- Make use of interviews, particularly early during prosecution, to save time in resolving any potential misunderstandings between the examiner and applicant, and to help save the examiner time in learning about the features of the invention and differentiation from the prior art.
- Before conducting an interview, prepare a complete yet concise agenda to help the examiner rapidly prepare for and to allow for efficient use of time allocated to the interview.

Patent applicants can help clear the pathway from patent application to granted patent by aligning aspects of their prosecution strategy with expectations placed on the examiner under the count system.

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[1] Munger, C. The Psychology of Human Misjudgment, 2005 revision, available at FS.Blog <https://fs.blog/great-talks/psychology-human-misjudgment/>.

[2] United States Patent and Trademark Office (USPTO), Examination Time and the Production System ("USPTO Production System Slides"), Slide 10.
<https://www.uspto.gov/sites/default/files/documents/Examination%20Time%20and%20the%20Production%20System.pdf>.

[3] USPTO, Patent Quality Assurance, <https://www.uspto.gov/patents/laws/examination-policy/patent-quality-assurance>.

[4] U.S. Office of Personnel Management, General Schedule Overview,
<https://www.opm.gov/policy-data-oversight/pay-leave/pay-systems/general-schedule/>.

[5] USPTO, Become a patent examiner,
<https://www.uspto.gov/sites/default/files/documents/USPTOPatentExaminerInfographicAPP LYNOW1.pdf>.

[6] Simmons, N.R., Putting Yourself in the Shoes of a Patent Examiner: Overview of the United States Patent and Trademark Office (USPTO) Patent Examiner Production (Count) System, 17 J. MARSHALL REV. INTELL. PROP. L. 32, 35 (2017).

[7] Id. and M.P.E.P, §1005; See also, Frakes, M.D. & Wasserman, M, Is the Time Allocated to Review Patent Applications Inducing Examiners to Grant Invalid Patents? Evidence from Micro-Level Application Data, Review of Economics and Statistics (2016) at 8-9, Available at: https://scholarship.law.duke.edu/faculty_scholarship/3646.

[8] See Simmons at 36.

[9] Id.

[10] USPTO, Work-life balance, <https://www.uspto.gov/jobs/why-choose-us/work-life-balance>.

[11] Supra, USPTO Production System Slides, Slide 14.

[12] Supra, USPTO Production System Slides, Slides 19 and 20.

[13] Supra, USPTO Production System Slides, Slides 16.

[14] Supra, USPTO Production System Slides.

[15] USPTO, Updates to Examination Time, Application Routing, and Examiner Performance Appraisals, 2019, <https://www.uspto.gov/patents/laws/examination-policy/updates-patent-examination-time-application-routing> and <https://www.uspto.gov/sites/default/files/documents/ExaminationProcessUpdatesMay2019.pptx>.

[16] In addition to different grade levels, Examiners can also be promoted to different steps, which are related to experience over time. Both grade level promotions (merit-based) and step promotions (experience-based) include salary increases. In this regard, grade level does not necessary correlate directly to years of experience; See GS Salary Table for 2022, <https://www.opm.gov/policy-data-oversight/pay-leave/salaries-wages/salary-tables/pdf/2022/GS.pdf>.

[17] USPTO Production System Slides, Slide 12.

[18] Tu, Shine (Sean), Three New Metrics for Patent Examiner Activity: Office Actions per Grant Ratio (OGR), Office Actions per Disposal Ratio (ODR), and Grant to Examiner Ratio (GER) (January 11, 2018). WVU College of Law Research Paper No. 2019-003, 100 Journal of the Patent and Trademark Office Society 277 (2018), <http://dx.doi.org/10.2139/ssrn.3100326>.

[19] USPTO, Patent Examiner Count System, <https://www.uspto.gov/patents/initiatives/patent-examiner-count-system#heading-3>; and USPTO Joint Labor and Management Count System Task Force, Overview of Count System Initiatives and Changes ("USPTO Count System Changes Slides"), Slide 7, Available at: https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.uspto.gov%2Fsites%2Fdefault%2Ffiles%2Fpatents%2Finit_events%2FCount_System_changes-Overview_3-8-2010.ppt&wdOrigin=BROWSELINK.

[20] USPTO Count System Changes Slides, Slide 17.

[21] Id.; See also USPTO Production System Slides, Slide 20, including an Example contemplating production in excess of 100%.

[22] The "rational maximizer" is a hypothetical person from rational-choice theory who makes decisions designed to achieve maximum preference satisfaction. When considering an employee evaluation system from the perspective of a rational employee, achieving maximum preference satisfaction can be defined as maximizing monetary compensation. While acknowledging that an individual human is complex, unique, and, not consistently rational (as is frequently discussed in connection with behavioral economics), the perspective of the rational maximizer remains useful when thinking about incentives created by systems applied to groups of people. For a discussion of rational-choice theory applied to systems governing groups of people, i.e., legal systems, see Richard A. Posner, "Rational Choice, Behavioral Economics, and the Law," 50 Stanford Law Review 1551 (1997).